

WASP-135b

R-1305

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210821 · created 2026-07-31T17:55:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459447.7423 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.027
Transit depth (Rp/Rs)^2	1.81 +/- 0.74 [%]
Orbital Inclination (inc)	84.57 +/- 1.83
Airmass coefficient 1 (a1)	1.121 +/- 0.036
Airmass coefficient 2 (a2)	-0.089 +/- 0.019
Scatter in the residuals of the lightcurve fit is	3.16 %
Best Comparison Star	#2 - [98, 72]
Optimal Aperture	2.88
Optimal Annulus	10.78
Transit Duration (day)	0.0806 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.027 vs 0.139 (deviation 0.12σ)
Duration fit vs published	0.0806 d vs 0.0687 d (deviation 1.16σ)
Mid-time O–C	5.5 min
Depth SNR	4.2
Residual scatter	3.16 %

Light curve

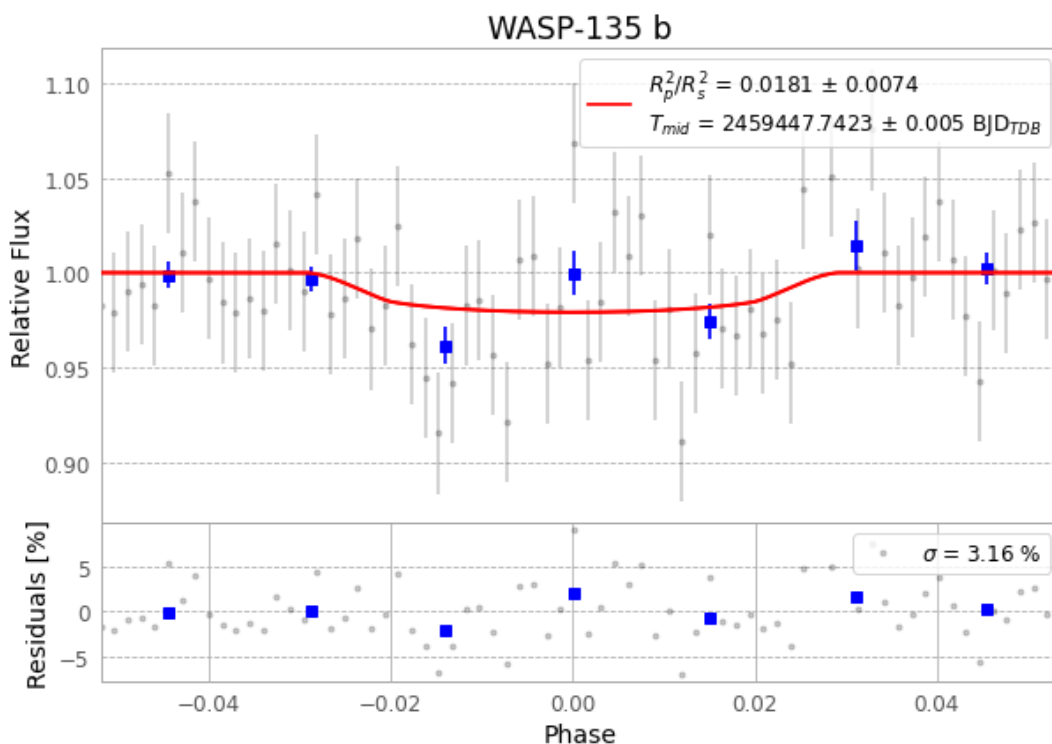


Figure 1 — FinalLightCurve_WASP-135 b_2021-08-20.png (SHA-256 db1dbcf6b433361e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [393, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2538527df3e8a151...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), WASP-135210821040012.FITS → WASP-135210821073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210821.log (a95f38889ec0...), FinalLightCurve_WASP-135 b_2021-08-20.png (db1dbcf6b433...), FinalLightCurve_WASP-135 b_2021-08-20.csv (71e41cbfe7dd...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:55Z.